

Generation Interconnection System Impact Study Report

For Queue Project Q17-0861

College And Walnut 138 kV

August 20, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0861. The Interconnection Customer proposes a 77.8 MW solar photovoltaic facility interconnecting at the College And Walnut 138 kV substation in the Pennsylvania area. The request was received on September 11, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 266 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$66,409,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of TALENENERGY MARTINS CREEK LLC HARRISBURG 500-345 kV TX, the FITZPATRICK-CENTRAL BEACH 138 KV loads to 183.8% of its 318 MVA rating. Project DFAX: 11.3%.

The CENTRAL BEACH-PAXON SUBSTATION 138 KV Line is overloaded to 182.3% of its 313 MVA emergency rating for the loss of the TALENENERGY MARTINS CRK LLC HARRISBURG 500-345 kV Auto. The project contributes 11.3% DFAX (8.8 MW) to this overload.

S LAKE GENEVA-PAXON 138 KV: 161.3% loading (206 MVA) following the outage of the TALENENERGY MARTINS CRK LLC HARRISBURG 500-345 kV XFMR; DFAX = 6.4%.

For the loss of TALENENERGY MARTINS CRK LLC HARRISBURG 500/345kV TX, the TALENENERGY MARTINS CRK LLC HARRISBURG-FITZPATRICK 138 KV loads to 161.3% of its 382 MVA rating. Project DFAX: 11.3%.

The KANSAS RIV PROJECT-METALCAN 138 KV is overloaded to 153.2% of its 304 MVA emergency rating for the loss of the TALENENERGY MARTINS CRK LLC HARRISBURG 500-345 kV TX. The project contributes 21.8% DFAX (17.0 MW) to this overload.

COLUMBUS-ZACHARY 138 KV Line: 148.4% loading (298 MVA) following the outage of the TALENENERGY MARTINS CREEK LLC HARRISBURG 500/345 kV Transformer; DFAX = 21.8%.

For the loss of TALENENERGY MARTINS CRK LLC HARRISBURG 500/345kV XFMR, the VICTORIA TC 7001-7007-7008-HAT CREEK 2 138 KV loads to 143.7% of its 230 MVA rating. Project DFAX: 6.9%.

KANSAS RIV PROJECT-COLUMBUS 138 KV Line: 143.4% loading (314 MVA) following the outage of the TALENENERGY MARTINS CRK LLC HARRISBURG 500/345kV Autotransformer; DFAX = 21.8%.

For the loss of TALENENERGY MARTINS CRK LLC HARRISBURG 500/345kV Auto, the BARNES LAKE-VICTORIA TC 7001-7007-7008 138 KV loads to 140.2% of its 252 MVA rating. Project DFAX: 6.9%.

BARNES LAKE-GRASS CRK 138 KV: 138.3% loading (262 MVA) following the outage of the TALENENERGY MARTINS CREEK LLC HARRISBURG 500/345kV Transformer Bank; DFAX = 6.9%.

For the loss of TALENENERGY MARTINS CRK LLC HARRISBURG SUBSTATION 500/345kV TX, the SAN LORENZO-GRASS CRK 138 KV loads to 136.6% of its 271 MVA rating. Project DFAX: 6.9%.

The SAN LORENZO-AJ SWOPE 138 KV is overloaded to 134.2% of its 286 MVA emergency rating for the loss of the TALENENERGY MARTINS CRK LLC HARRISBURG 500/345kV Transformer. The project contributes 6.9% DFAX

(5.4 MW) to this overload.

The KNOXVILLE BATTERY ENERGY STORAGE-HAMILTON PA 138 KV is overloaded to 133.5% of its 166 MVA emergency rating for the loss of the ZACHARY-LEVEE 138 KV Line. The project contributes 16.7% DFAX (13.0 MW) to this overload.

For the loss of TALENENERGY MARTINS CRK LLC HARRISBURG 500 kV/345 kV Transformer Bank, the LEVEE-ZACHARY 138 KV line loads to 131.3% of its 215 MVA rating. Project DFAX: 8.5%.

For the loss of TALENENERGY MARTINS CRK LLC HARRISBURG 500/345kV Auto, the SOUTH LAKE GENEVA-HUMMELSTOWN 138 KV loads to 131.2% of its 230 MVA rating. Project DFAX: 6.4%.

TALENENERGY MARTINS CRK LLC WEST SHORE-AJ SWOPE 138 KV L: 131.0% loading (306 MVA) following the outage of the TALENENERGY MARTINS CRK LLC HARRISBURG 500-345 kV TX; DFAX = 6.9%.

HAT CRK 2-HARGROVE ROAD D.S 138 KV line: 130.7% loading (241 MVA) following the outage of the TALENENERGY MARTINS CRK LLC HARRISBURG 500/345kV Transformer Bank; DFAX = 6.9%.

For the loss of TALENENERGY MARTINS CRK LLC HARRISBURG 500/345kV Transformer, the RIVER BEND-HUMMELSTOWN 138 KV loads to 121.1% of its 239 MVA rating. Project DFAX: 6.4%.

For the loss of TALENENERGY MARTINS CRK LLC HARRISBURG 500/345 kV XFMR, the RIVER BEND-GARBERVILLE 138 KV loads to 111.8% of its 248 MVA rating. Project DFAX: 6.4%.

COLLEGE AND WALNUT-UKIAH 138 KV: 110.4% loading (255 MVA) following the outage of the CARDWELL-BEMIDJI SUBSTATION 138 KV; DFAX = 24.8%.

OTTEN-KNOXVILLE BATTERY ENERGY STORAGE 138 KV: 108.6% loading (194 MVA) following the outage of the ZACHARY SUBSTATION-LEVEE 138 KV; DFAX = 16.7%.

The UKIAH-METALCAN 138 KV line is overloaded to 107.7% of its 250 MVA emergency rating for the loss of the BEMIDJI-CARDWELL 138 KV. The project contributes 24.8% DFAX (19.3 MW) to this overload.

ZACHARY-HAMILTON PA 138 KV: 106.4% loading (208 MVA) following the outage of the LEVEE-ZACHARY 138 KV; DFAX = 16.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FITZPATRICK-CENTRAL BEACH 138 KV	Rebuild line with higher-capacity conductor	\$1,578,000	38% (\$598,000)
CENTRAL BEACH-PAXON SUBSTATION 138 KV Line	Rebuild line with higher-capacity conductor	\$3,444,000	38% (\$1,305,000)
S LAKE GENEVA-PAXON 138 KV	Rebuild line with higher-capacity conductor	\$1,162,000	100% (\$1,162,000)
TALENENERGY MARTINS CRK LLC HARRISBURG-FITZPATRICK 138 KV	Rebuild line with higher-capacity conductor	\$1,871,000	38% (\$709,000)
METALCAN-KANSAS RIVER PROJECT 138 KV	Rebuild line with higher-capacity conductor	\$2,268,000	100% (\$2,268,000)
COLUMBUS-ZACHARY 138 KV	Rebuild line with higher-capacity conductor	\$3,535,000	100% (\$3,535,000)
VICTORIA TC 7001-7007-7008 SUBSTATION-HAT CRK 2 SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$2,477,000	100% (\$2,477,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
KANSAS RIV PROJECT-COLUMBUS 138 KV Line	Rebuild line with higher-capacity conductor	\$5,009,000	100% (\$5,009,000)
BARNES LAKE-VICTORIA TC 7001-7007-7008 138 KV	Rebuild line with higher-capacity conductor	\$4,110,000	100% (\$4,110,000)
BARNES LAKE-GRASS CRK 138 KV	Rebuild line with higher-capacity conductor	\$3,364,000	100% (\$3,364,000)
SAN LORENZO-GRASS CRK 138 KV	Rebuild line with higher-capacity conductor	\$2,616,000	100% (\$2,616,000)
SAN LORENZO-AJ SWOPE 138 KV	Rebuild line with higher-capacity conductor	\$3,817,000	100% (\$3,817,000)
HAMILTON PA-KNOXVILLE BATTERY ENERGY STORAGE 138 KV	Rebuild line with higher-capacity conductor	\$2,622,000	100% (\$2,622,000)
LEVEE-ZACHARY 138 KV	Rebuild line with higher-capacity conductor	\$19,667,000	43% (\$8,384,000)
SOUTH LAKE GENEVA-HUMMELSTOWN 138 KV	Rebuild line with higher-capacity conductor	\$4,393,000	100% (\$4,393,000)
AJ SWOPE-TALENENERGY MARTINS CRK LLC W SHORE 138 KV	Rebuild line with higher-capacity conductor	\$6,770,000	100% (\$6,770,000)
HAT CRK 2-HARGROVE ROAD D.S 138 KV line	Rebuild line with higher-capacity conductor	\$1,370,000	100% (\$1,370,000)
RIVER BEND-HUMMELSTOWN 138 KV	Rebuild line with higher-capacity conductor	\$2,044,000	100% (\$2,044,000)
RIVER BEND-GARBERVILLE 138 KV	Rebuild line with higher-capacity conductor	\$2,264,000	100% (\$2,264,000)
COLLEGE AND WALNUT-UKIAH 138 KV	Rebuild line with higher-capacity conductor	\$4,105,000	100% (\$4,105,000)
OTTEN-KNOXVILLE BATTERY ENERGY STORAGE 138 KV	Reconductor line	\$709,000	100% (\$709,000)
UKIAH-METALCAN 138 KV line	Reconductor line	\$1,770,000	100% (\$1,770,000)
HAMILTON PA-ZACHARY 138 KV line	Reconductor line	\$1,008,000	100% (\$1,008,000)
	Total allocated to project		\$66,409,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.